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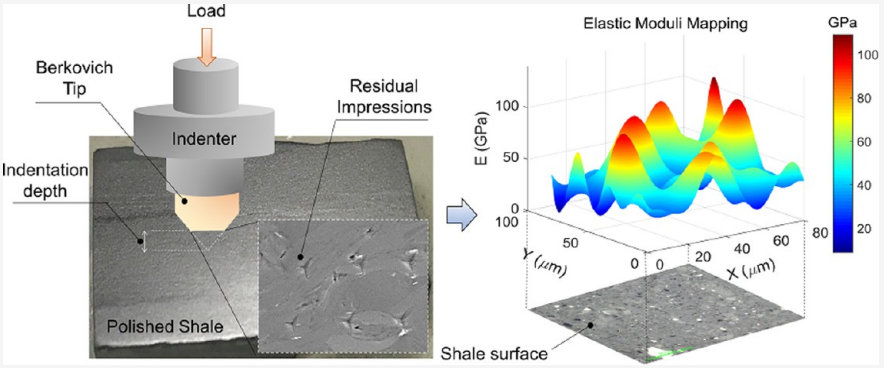
Probing Nanomechanical Properties of a Shale with Nanoindentation: Heterogeneity and the Effect of Water–Shale Interactions

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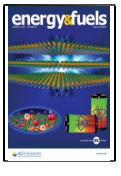


ABSTRACT: The mechanical properties of shale are critical for the extraction of gas and oil from nanoporous shale. In this study, nanoindentation tests with the accelerated property mapping mode were used to obtain the localized quantitative mechanical properties (the elastic modulus and hardness) of the tested shale sample. X-ray diffraction was used to analyze the bulk mineral composition. The Gaussian mixture model and k-means algorithm were used to fit the data and identify distinct mineral phases based on load–displacement curves. Three clusters of different mineral phases were recognized through nanoindentation curves. Experimental results and analyses show that the nanomechanics of shale are conditioned by the mineral composition and its spatial distribution. The spatial distribution of the elastic modulus and hardness showed the high heterogeneity of shale at the nano-to-micron scale, closely related to its localized mineralogical distribution and microstructures. It was found that the plastic work induced in the nanoindentation process was different between the three mineral clusters with a plastic work ratio of 0.06–0.18, 0.32–0.38, and 0.46–0.53 in high-, medium- and low-hardness clusters, respectively. A two-term exponential relationship was observed between hardness and plastic work. For the tested Marcellus shale, obvious mechanical weakening was observed after 20 days of water vapor treatment. The elastic modulus of shale exhibited a reduction of 0.6–15%, and the hardness decreased by 8.6–17.8%. Further investigation is recommended to quantify the variations in its indent-specified mechanical properties and its direct relationship with localized mineralogy and microstructure.

1. INTRODUCTION

Shale is a thin-sheet sedimentary rock formed by the compaction of silt and clay particles as well as inserted organic matter and other inorganic minerals. With the development of advanced hydraulic fracturing and horizontal drilling technologies, unconventional resources, represented by tight oil and shale gas/oil, have become economically feasible energy sources.[1,2] Shale reservoir has low permeability, and stimulation treatments are required to achieve economical flowrates for it.[3] The mechanical response of shale plays an important role in nearly every stage of oil and gas extraction, and the water–shale interaction has effects on the borehole integrity and transport of hydrocarbons. Therefore, a comprehensive understanding of the mechanical properties of shale is crucial to its exploitation in the industry. The mechanical parameters of rocks, such as the elastic modulus, hardness, and fracture toughness, have a great influence on the geometry of the fracture network, impacting the proppant embedment and permeability of the stimulated formations.[4] In the past, the most common ways to evaluate the mechanical properties of shale and the degree of the water–shale

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interaction are the conventional mechanical tests, including uniaxial and triaxial compressive strength tests, shear tests, and ultrasonic measurements.[5−14] These meso- or macroscopic laboratory tests can provide only the average value of the all-solid phase, which fail to consider the microstructures' heterogeneity and the mineralogical compositions.[15−21] In fact, the length scale involved in the production of shale gas can span up to 13 orders of magnitude.[22] The mechanical characterization of shale on a macroscale is not enough for engineering applications. Thus, it is necessary to conduct detailed laboratory experiments down to the submicron scale to obtain the mechanical property of shale and its alteration after the water−shale interaction.

In view of the inherent heterogeneity and anisotropy of the geomaterials, the characterization of its mechanical properties at the nanoscale is necessary for the multiscale modeling of shale. Former studies reported that mechanical properties of rocks at the submicron scale can be determined by nanoindentation techniques, such as Young's modulus, hardness, fracture toughness, and creep behavior.[23−25] Previous investigations found that the mechanical responses of the rock-forming minerals are significantly different.[26,27] These minerals demonstrate clear elasticity- or plastic-preferred characteristics to the load and show various degrees of irrecoverable deformation.[27−30] The mechanical differences among the minerals, either relatively hard granular particles or soft kerogens, have been shown to have a crucial impact on the bulk mechanical properties of shale.[31,32] In addition, it is found that the mechanical properties such as its strength, stiffness, and fracture roughness are orientation-dependent, and the multiscale anisotropy of shale is attributed to mechanisms corresponding to the observed scale.[4,30,33−35] For the purpose of ultimate application in the industry, upscaling the nano- or micromechanical properties to the engineering scale is a key step. Although still in infancy, some studies have tried to upscale the nanoscale strength of shale rock to bulk strength by homogenizing the multiphase elastic composites using the Mori−Tanaka model.[27,36−39]

To date, studies on the nanoscale characterization of shale have been mainly based on the nanoindentation test, also called instrument indentation test. Due to the improvement in the sensor technology and high-resolution imaging, the instrument indentation test can characterize the properties of materials at the submicron scale and thus has been routinely performed as a supplement to regular mechanical tests.[40−43] It has been employed in a broad spectrum of materials, such as thin films,[44] polymers,[45] superalloys,[46] biomaterials,[47] and rocks.[48] In a nanoindentation test, a probe of known geometry (pyramid, sphere, or cubic) penetrates the surface of the sample under a given load until an impression is formed. By recording the forces and the corresponding depths during the test, the elastic modulus and hardness of the tested material can be determined. This offers a way for the quantitative mechanical characterization of highly localized heterogeneous materials.[49] Notably, the flexibility in the shape and size of the tested sample makes the nanoindentation test easier and faster. A small volume of rock chips or fragments is enough for the test, and no large core or bulk sample is needed. Moreover, the nanoindentation test is nondestructive and friendly to brittle rock. Combined with scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS), the mineralogy at the indentation site can be investigated. Apart from the nanoindentation test, the atomic force microscopy (AFM) is another commonly used technique to characterize nanoscale mechanical properties in recent years.[50−52] It can provide the quantitative mechanical map and three-dimensional (3D) surface profile of the tested material.[31] However, the AFM technique can yield large testing errors when measuring high-modulus materials.[53] Nanoindentation is therefore a good choice for characterizing mechanical properties at the nanoscale.

Our knowledge of the mechanical properties of shale at the nanoscale is limited because of the limited amount of available data. Understanding how the mineralogy and microstructure influence the deformation- and transport-related behavior of shale is critical for bottom-up methods to explain the observed bulk behavior in the lab. This can serve as the stepping stone for the cross-scale data integration and predictive physics modeling of the mechanical and transport behaviors of shale. Besides, the mechanical alteration of shale after water sorption lacks experimental evidence on a nanoscale. In this study, a series of nanoindentation tests were conducted on the shale sample collected from the Marcellus formation, obtaining the elastic modulus and hardness of the sample at the local submicron area. Statistical methods were used to analyze the indentation data and figure out the mineralogy-related mechanical behaviors of shale. The mechanical alteration after water−shale interaction was investigated. This nanoscale characterization method for shale provided a simple and nondestructive way of quantitative appraisal of the mechanical response of shale at the nanoscale.

2. NANOSCALE MECHANICAL PROPERTIES OF SHALE AND THE INFLUENCING FACTORS

2.1. Mineralogy of Shale at the Nanoscale. Mineralogy is a key parameter that impacts the micromechanical properties of rocks. There is a wide range of minerals presented in shale. Previous studies transformed the petrophysical model of shale into a three-component mechanical model, which classified the constituent minerals of shale into three groups because of their similar material properties: stiff minerals like quartz, medium-strength minerals, and compliant parts like organic matter and clay.[4,54−56] In a previous study on the nanocharacteristics of shale samples, distinct differences were noted between mineral grains in terms of strength and stiffness.[30] Silicates with a framework crystal structure (tectosilicates), such as quartz and feldspar, exhibit the highest strength and stiffness, followed by phyllosilicates like the clay phase. Organic matter, as the relatively soft part in shale, exhibits compliant mechanical behaviors to loads. Previous studies reported an overall negative correlation between Young's modulus and the clay content, total organic carbon (TOC), and porosity, whereas a positive correlation has been noted between the contents of carbonate/quartz and Young's modulus based on experimental data from different shales.[26,57] The distinct elastic−plastic responses of the minerals result in a diverse set of mechanical properties.[27−30] The micrometer-sized clay minerals play a vital role in the mechanical behaviors of shale due to their orientation-preferred stacking arrangements and hydration characteristics, which depend on the structure units that form the clay minerals. Nonclay minerals in shale, such as quartz and feldspar, are formed by different dense crystal structures that may be attributed to the distinct elastic−plastic behaviors among these mineral grains.[28] And organic matters, the final possible solid component of shale, usually with an amorphous granular structure or dispersed in the matrix, also

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strongly affect the deformation of shale. Besides, the mineral compositions influence the microstructure heterogeneities and lead to various mechanical behaviors of the rock. The mineralogy and nanopore structure directly impact the water sorption and retention behavior of shale, which is ascribed to the capillary action of the shale containing nonswelling or swelling clays.[16] Previous studies suggest that the mineralogical compositions strongly influence the microstructure of shale. Quartz contents and TOC are positively related to the specific surface area and total pore volume, whereas clay mineral contents are negatively related to the specific surface area and total pore volume.[58,59] The pore structures and microstructure system on nanoscale are crucial for the transportation of hydrocarbon fluids. They also provide a hydration space and may lead to the destruction of the integrity as well as mechanical weakening. Nonetheless, the ways in which the mineralogy and the microstructure can intrinsically determine its overall behavior should be further investigated.

2.2. Water−Shale Interaction and Its Weakening Effect on Mechanical Properties. Water is omnipresent in the subsurface fluid-bearing formations. It can trigger the structural modification of shale and deteriorate its integrity.[21,56,60,61] Mechanical tests on rocks in the laboratory have shown that their mechanical properties weaken after exposure to water or moisture, including reductions in the strength and stiffness, augmentation of creep behavior, and plastic deformation as well as accelerated failure rates.[62−67] Due to variations in the minerology and multiscale heterogeneity of shales, it is challenging to quantify the fundamental mechanism of their deterioration with water uptake. Potential mechanisms of the water-weakening effect on shale have been conceptualized and proposed to explain the deterioration of its integrity. The water-induced mechanical softening of shale may be attributed to multiple factors, including the chemical and corrosive deterioration of rocks, increase in pore pressure, and reductions in capillary tension and interface cohesive strength.[64,68,69] Reactions between the water and diagenetic minerals such as calcite and silica in shale and their dissolution may increase the porosity of the rock, as illustrated in Figure 1a.[70] Moreover, the crystalline swelling from water-sensitive clay minerals has been shown to be a dominant factor that influences the reduction in the strength of shale after water soaking by altering the microstructure of clayey rocks, as shown in Figure 1c.[63,71] With water uptake, clay minerals with interlayers, like smectite, have a tendency to hydrate exchangeable interlayer cations to force the clay layers apart.[72] For different types of clay minerals, the hydration rate and hydration stress vary significantly. The unsynchronized expansions among clay minerals result in nonuniform stress, cause local stress concentration, and further drive the initiation of microcracks in shale and lead to a reduction in strength (Figure 1c).[14] Beyond that, water can infiltrate the intergranular boundaries and weaken the cementation between the mineral particles, thereby reducing the cohesive strength. This can manifest itself as a decrease in cohesiveness, frictional angle, and the strength of the rock at the macroscale, as shown in Figure 1b.[14,68] In addition, past work has shown that capillary pressure in the water−gas interface and the resulting suction can significantly impact the mechanical properties of shale.[73] At a low degree of saturation, water fills the small pores or narrow spaces between grains to generate high capillary suction inside the shale. This acts as confining pressure and leads to the closure of pre-existing microcracks, explaining the

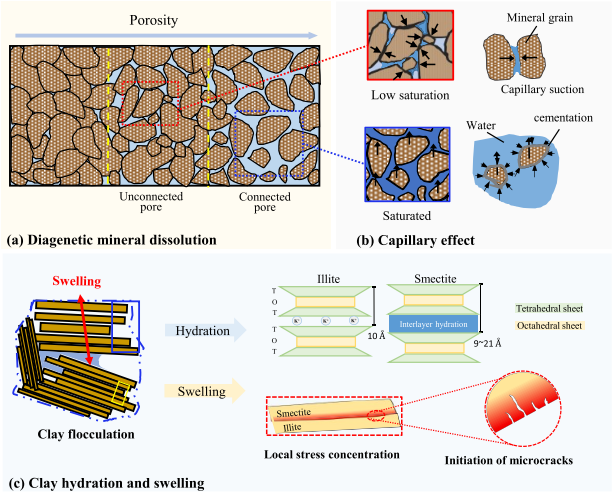


Figure 1. Schematic of potential water−shale interactions and the weakening effects. (a) Water uptake induces the dissolution of diagenetic minerals. (b) Water can weaken the cementation between mineral particles and reduce cohesive strength. (c) The nonuniform swelling of clay minerals may provoke the propagation and further connection of the cracks to reduce strength and stiffness and ultimately lead to a failure.

increase in strength early on. As the water content increases, the confining effect due to suction is released, and water penetrates the reopened microcracks and reduces the strength of the rock, as shown in Figure 1b.

3. EXPERIMENTAL WORK

3.1. Sample and Specimen Preparation. The shale core was drilled from the Marcellus formation near the Allegheny Structural Front (ASF) in the middle of Pennsylvania that is marked by the red star in Figure 2. It was cut into small cubes paralleled to the bedding

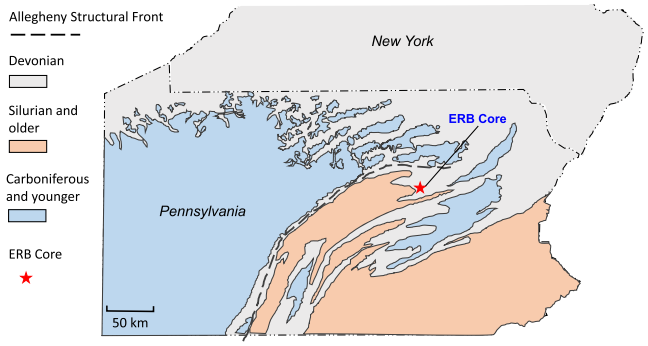


Figure 2. Geographical map of Allegheny Structural Front (ASF) and the location of the shale.[74]

plane direction for the nanoindentation test. To eliminate the impact of the adsorbed water, the sample was oven-dried for 24 h at 80 °C and sealed in a sample box. A flat surface is required in the nanoindentation test for the accuracy of the results. Thus, careful polishing was conducted. Prior to the polishing procedure, the shale sample was casted in epoxy resin that consists of the epoxy resin and hardener by weight of 5:1 to prevent the sample from spalling or damaging during the surface preparation. The mixture resin was cast into a prelubricated 1.5-in. diameter cylindrical casting mold. After the epoxy resin fully cured, the sample was fixed on a metal specimen disk with crystal bond to facilitate the polishing. Grinding and polishing were performed on the MultiPrep Surface Preparation Machine (Allied High-Tech Products, Inc.) with abrasive papers of different

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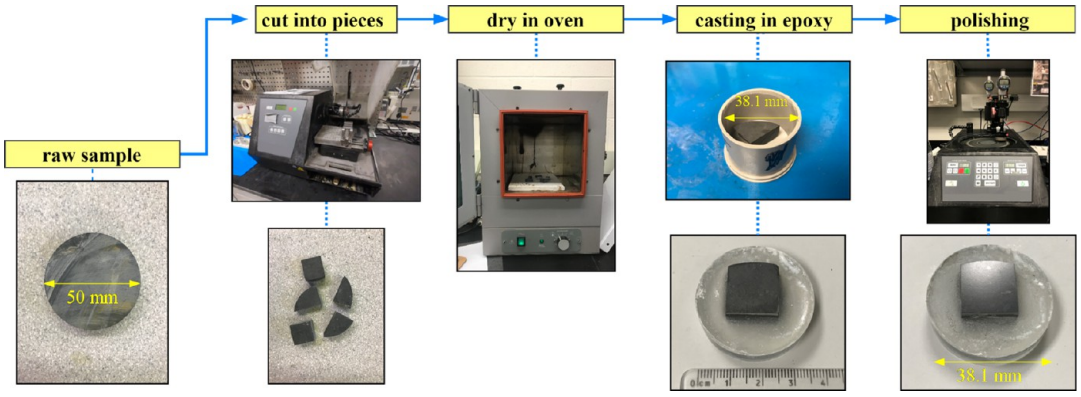


Figure 3. Sample preparation procedure for nanoindentation testing.

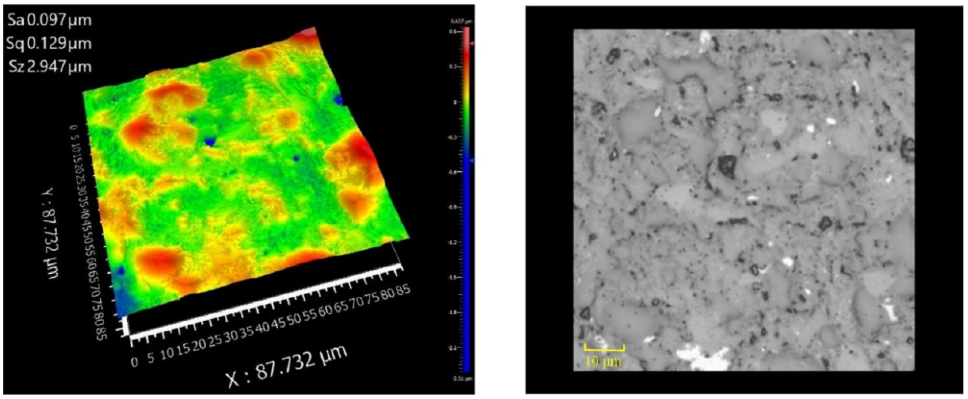


Figure 4. Surface topography of the sample after fine polishing shows that the surface roughness is 129 nm as measured by an optical profilometer.

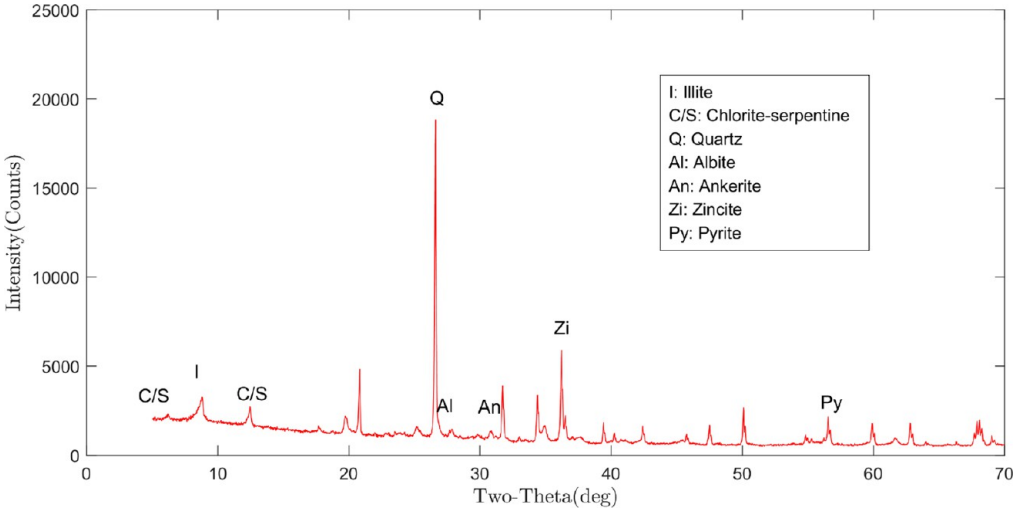


Figure 5. X-ray diffraction pattern of the studied shale sample where zincite was added to the sample powder for quantitative compositional analysis.

grain fineness levels of grit 320, 600, 800, and 1200 (*i.e.*, P400, P800, P1200, and P4000, with equivalent grit sizes of 35, 22, 15, 10, and 5 μm, respectively). Then, the sample was further polished to reach a finer surface using 1 μm, 0.25 μm polycrystalline diamond suspension followed by 0.05 μm alumina suspension. The entire preparation procedure is depicted in Figure 3. During the whole polishing process, we used water-free, alcohol-based lubricants and ethanol and ethylene- or glycol-based suspension to avoid potential mineral hydrations. Upon the completion of surface polishing, ethanol was used to clean the surface of the sample and pressurized filtered air was used to dry the surface for the purpose of a clean and smooth surface.

Surface roughness is very important in the indentation measurement. It can significantly impact the accuracy of the indentation results by causing errors in the determination of the projected area of the probe and the tested sample.[75] Thus, it is critical to ensure that the indentation tests are performed on a smooth surface. A previous study recommended that for homogeneous materials such as metals, the indentation depth should be 20 times larger than the roughness average (Ra), while for porous media like bones, the indentation depth that is 3 times larger than the root mean squared (RMS) roughness can avoid the impact of roughness.[76,77] In this work, the RMS roughness of the polished sample can reach approximately 129

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nm (as shown in Figure 4), which is measured by an optical profilometer. In view of the presence of pores on the surface, the actual roughness is smaller than 129 nm, and the risk of inaccuracy can be further decreased by carefully selecting the tested area with less pores and flaws.

3.2. X-ray Diffraction (XRD) Analysis of the Sample. XRD was conducted to obtain the mineralogical compositions using a RIGAKUD/Max-3B diffractometer at Penn State Material Characterization Lab (PSU-MCL). The raw diffraction data were interpreted by the Jade software to determine the mineral contents. The collected shale chips were crushed to 325 mesh powder, and the scan angle ranges in the XRD were from 5 to 70° under vacuum and at room temperature. The relevant signal is shown in Figure 5. The mineral composition analysis result based on XRD analysis is summarized in Table 1. The shale sample is mainly comprised of quartz, illite, chlorite-serpentine, albite, pyrite, and ankerite. The sample is clay-rich and contains illite with a weight of 39.8%. The second most common composition mineral is quartz, with a weight fraction of 30.2%. The contents of carbonate and feldspar are relatively minor, with 2.8% ankerite and 8.2% albite. The content of pyrite is significant, with a weight of 3.3%. The two-component interstratified mineral chlorite-serpentine was found in the sample with a weight fraction of 15.7%.

Table 1. Mineral Composition of the ERB Shale Sample Based on XRD Analysis

minerals	quartz	albite	ankerite	illite	pyrite	chlorite-serpentine
weight fraction (%)	30.2	8.2	2.8	39.8	3.3	15.7

3.3. Experimental Method of Nanoindentation Tests. The Berkovich probe was used for nanoindentation tests. It is a three-side pyramid, as shown is Figure 6a. A typical load−displacement curve is shown in Figure 6b, which consists of three segments: a linear loading process to the maximum preset force, a holding at the maximum force, and finally an unloading process back to zero load.

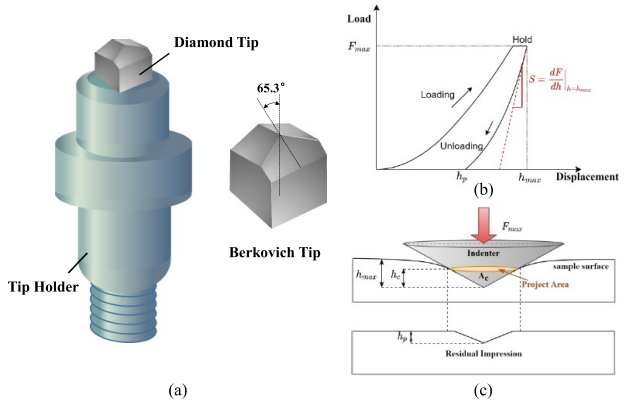


Figure 6. (a) Schematic of the indenter and Berkovich tip. (b) Typical load−displacement curve in the nanoindentation test using the three-segment load function. (c) Schematic of indenter and specimen surface at full load and after unloading process. F_{max} is the maximum load, h_{max} is the maximum deformation, and h_p is the residual impression after probe withdrawal from the specimen surface. A_c is the project area function.

The nanoindentation tests were conducted on the Hysitron TI 980 TriboIndenter (Bruker Corp., US), as shown in Figure 7a. The instrument has a load resolution of 1 nN and displacement resolution of 0.04 nm, with thermal drift less than 0.05 nm/s. An advanced accelerated property mapping mode was adopted to perform the indentations. The traditional nanoindentation measurement is extremely time-consuming because of the repeated location of each single indent and system calibration. Different with that, the accelerated property mapping mode is able to measure the nanomechanical properties of the material at six measurements per second by the coupling of a high-bandwidth electrostatically actuated transducer, fast control, and data acquisition electronics. Using both optical and scanning probe microscopy techniques, as shown in Figure 7b, accelerated property mapping mode can rapidly screen the surface and position features of interested regions and acquire statistical and spatial distributions of shale properties. A calibrated Berkovich intender tip was used with maximum load of 5 mN, as shown in Figure 7c. The typical three-segment load function was adopted, including 0.5 s loading, 0.2 s holding, and 0.5 s unloading. Eight sites were picked according to the local roughness, as schematically shown in Figure 7d. The number of indents is 64 for each site, with 10 μm spacing between two adjacent indents. Throughout the whole indentation process, the load−displacement curves were continuously monitored and recorded.

To investigate whether water vapor changes the mechanical properties of shale sample, we put the sample into a container with constant relative humidity of 85% for 10 and 20 days. Nanoindentation tests were conducted after water treatment on the region that is adjacent to the previous sites to ensure that the mineral characteristics are as close as possible. Pure potassium chloride powder (>99%) was used to generate a saturated salt solution and stabilized the 85% relative humidity in the container at 20 °C.[78] A relative humidity sensor was equipped in the environmental chamber for humidity monitoring, as shown in Figure 7e. The mechanical properties were compared before and after the water vapor treatment.

Nanoindentation is widely adopted in metallic materials, and the detailed theory is available in the literature if one is interested. In this work, the method proposed by Oliver and Pharr[42] was used to calculate Young's modulus and hardness from load−displacement curves, as follows:

$$E = (1-\nu^2)\left[\frac{1}{E_r} - \frac{1-\nu_i^2}{E_i}\right]^{-1} \quad (1)$$

$$H = \frac{F_{max}}{A_c} \quad (2)$$

where E, E_i, ν, and ν_i are the elastic modulus and Poisson's ratios of the test specimen and the indenter, respectively. E_r is the reduced modulus of the system, which can be calculated from the following:[42,75]

$$S = \frac{dF}{dh}\Big|_{h=h_{max}} = \frac{2\beta\sqrt{A_c}}{\sqrt{\pi}}E_r \quad (3)$$

where contact stiffness S is the slope of the unloading curve, β is a geometrical factor and $\beta = 1.034$, h is the displacement of the indenter tip, and h_{max} is the maximum depth at the maximum load. The project area function A_c is a function of the contact depth of the indenter and the surface of the material being tested h_c.

For the diamond tip we used in this work, E_i= 1140 GPa and ν_i=0.07. The Poisson's ratio of the shale sample generally varies between 0.2 and 0.3. Previous studies demonstrate that the value of elastic modulus is not very sensitive when the range of changes in Poisson's ratio is small.[23,79] A Poisson's ratio with 0.1 uncertainty only brings about 5% uncertainty in Young's modulus.[79] In this work, the Poisson's ratio of shale is assumed to be 0.25.

4. RESULTS AND ANALYSIS

4.1. Statistical Analysis of Load−Displacement Curves. A total of 512 indentations were performed on the sample. The load−displacement curves are shown in Figure 8. The curves of the eight sites show good consistency with the typical load−displacement curves as shown in Figure 6b. The three segments of the loading function were easily distinguishable on the curves. In contrast to the perfect load−

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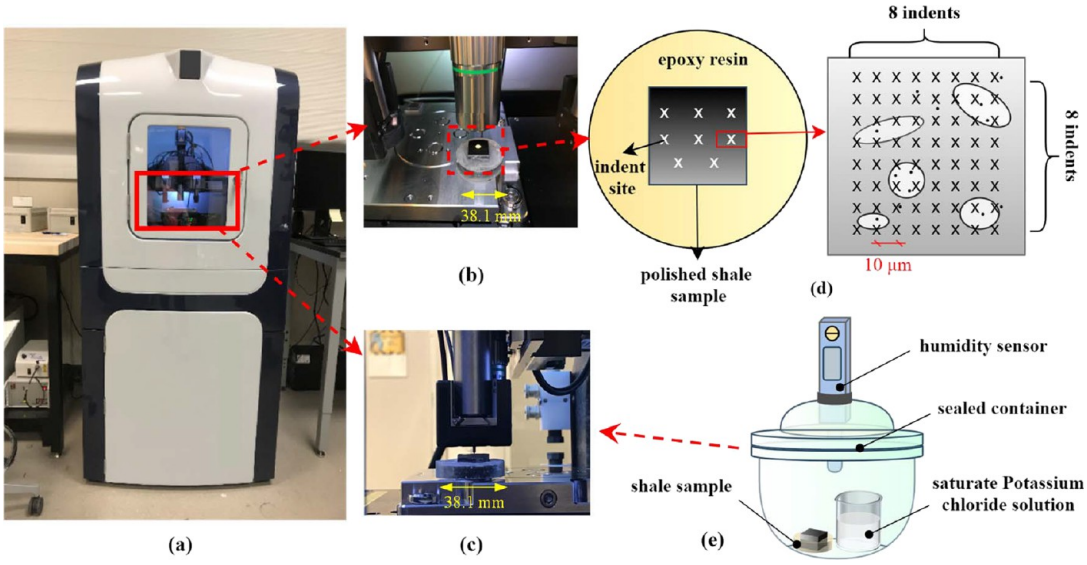


Figure 7. Schematic diagram of the nanoindentation experiment. (a) The nanoindenter that was adopted in the experiment. (b) The built-in optical microscope benefits the orientation of proper indentation site. (c) Local zoom of the indenter tip. (d) Eight sites were picked according to the local roughness in each nanoindentation test. On each site, there are 64 indents with 10 μm spacing. (e) The shale sample was placed into a container and treated with water vapor for up to 20 days.

displacement curves, sudden changes in slope were observed at the loading segment. These abnormal phenomena, also called pop-in events, demonstrate the characteristics of heterogeneity and the microstructure of shale, which will be discussed in the discussion.

Figure 8 shows that the maximum depth of a majority of the curves ranged from 160 to 400 nm, with a few over 500 nm. The elastic modulus and hardness of the local indented areas were extracted according to these curves. The elastic modulus E, calculated by eq 1, measures the nonpermanent deformation capability of the shale sample, manifested as the extent to which the indented surface returns to its previous shape. The hardness H, which depends on the resistance between the surface mineral and the localized deformation induced by the load of the indenter, can be calculated by eq 2. The shape of load−displacement curves contains rich information on the localized mechanical properties. The corresponding mineral of each load−displacement curves of the indents is unknown. Thus, statistical methods were used to analyze the data. Each curve was considered as an independent statistical sample with respect to the position of indentation. The experimental results exhibited a clear multimodal behavior in the histograms. Thus, it was assumed that the mechanical properties of each phase obeyed a Gaussian mixture distribution. We fitted a Gaussian mixture model (GMM) object to the indentation modulus data. Through the expectation maximization algorithm, the data were divided into three components of a Gaussian distribution, and the corresponding means, covariances, and mixing proportions were obtained. Each mixing proportion represented the fraction of the population described by a corresponding component. The three Gaussian distributions are depicted in Figure 9 as the purple, green, and yellow curves. The red curve in Figure 9 is the probability density function of the distribution.

The Gaussian mixture distribution of the elastic moduli of the 512 experimental indents for the eight sites is shown in Figure 9a. Previous studies revealed that the petrophysical model of shale can be transformed into a micromechanical parameter model that consists of three dominate mineral groups: high-, medium-, and low-strength minerals.[27,39] For different mineral groups, the mechanical responses to the indentation load were different, which manifest themselves as the different clusters in the mixture distribution. According to the statistical analysis, three clusters were recognized, with mean values of 35.3, 40.2, and 74 GPa, corresponding to the mean estimations of the three mineralogical phases of the shale sample. The results of the statistical analysis are summarized in Table 2. Interestingly, we found that the range of the mean elastic modulus of the clusters is comparable across each site, which confirms the statistically consistency. It indicates that the number of indents, although not very large, can reflect the micromechanical properties because shale is an ultrafine rock. The corresponding mixing ratios of the three mineral phases identified at different sites change with the location of indentation, indicating the heterogeneity of the shale sample. Combined with the XRD result, we infer that the cluster with the lowest strength most probably has soft minerals or phases distributed in it, such as clay matrix and organic matter. The medium-strength cluster may contain mineral grains with medium strength, like ankerite, inserted into a clay matrix, like albite, while the high-strength cluster is likely to be the hardest minerals, such as quartz and pyrite.

The surface morphology of the tested shale sample under the optical microscope is shown in Figure 10. It shows that the mineral grains are quite flat after being polished. A large area of the surface is occupied by nongranular minerals, which are mainly clay minerals as well as some kerogen, and it reflects the high proportion of the low-strength cluster identified in the statistical analysis. As shown in Table 2, the average value of the elastic moduli of the three strength clusters ranged from 30.9 to 39.9, 37.7 to 66.4, and 70 to 89.5 GPa from low to high strength, respectively. The average value of the lowest-strength cluster was slightly higher than the indentation modulus of the reported pure clay minerals (26−34 GPa) in previous studies.[28,51,80] This is because the mean value of the lowest-strength clusters was the statistical probability of the

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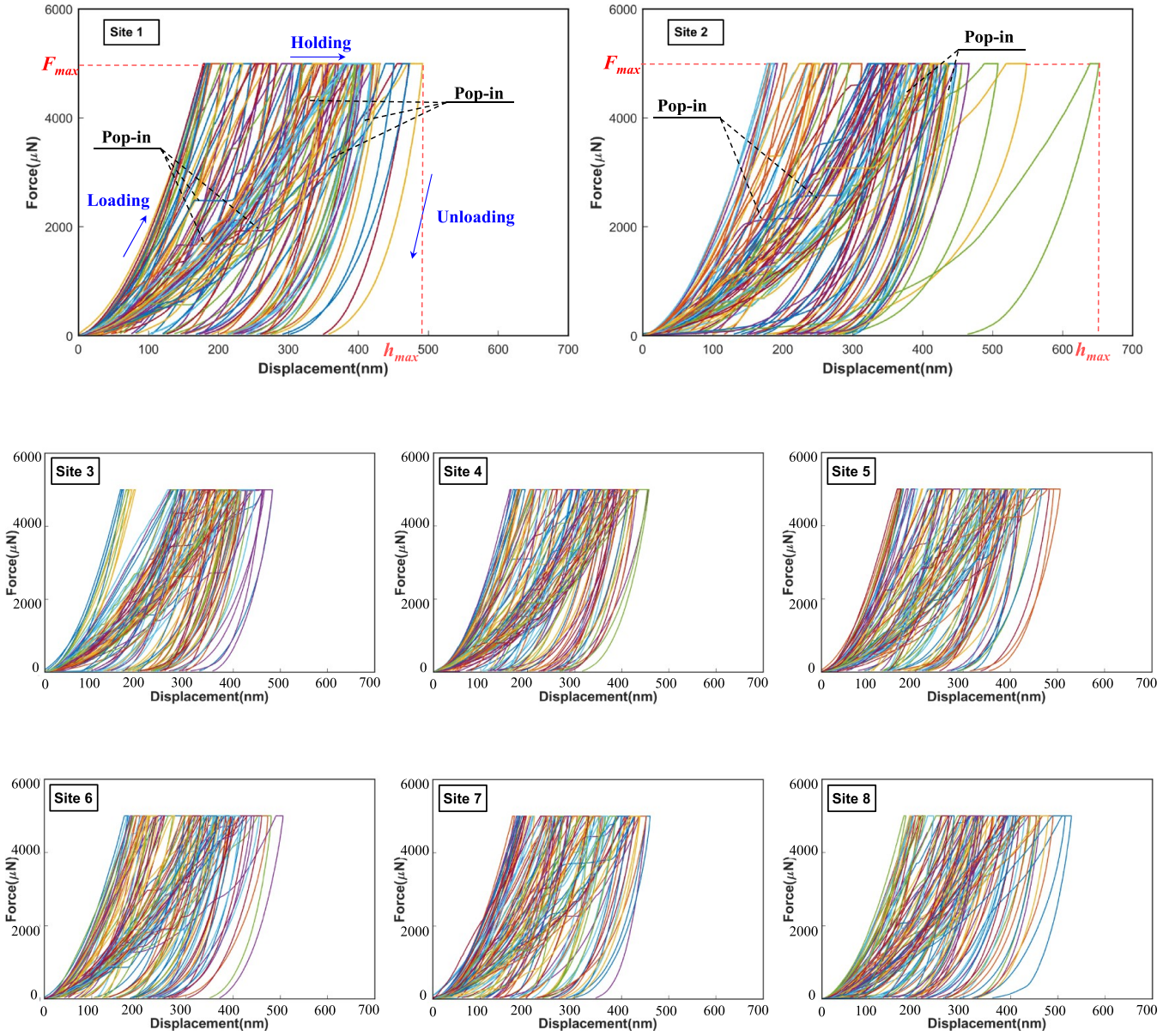


Figure 8. Load−displacement curves in the nanoindentation tests. The elastic modulus and hardness were extracted from these curves according to the slope of the unloading curve and the maximum deformation h_{max} at the maximum load F_{max}.

mechanical properties not only of clay minerals but also other tiny or broken granular minerals embedded in the clay matrix. Meanwhile, the mineralogy on the substrate of the indented area also has an influence on the tested value of the surface minerals, which may increase the indentation modulus. The mean elastic moduli of the medium-strength cluster, which may be attributed to the albite and ankerite minerals in shale, were consistent with the indentation moduli of feldspar reported in the literature, with a spectrum from 45 to 68 GPa,[80,81] and carbonates, ranging from 34 to 59 GPa.[51,52,81] For the highest-modulus cluster, the elastic modulus was up to 120 GPa, indicating that the hard cluster can be attributed to the hard minerals, like quartz and pyrite, and the tested moduli were consistent with previously reported indentation moduli.[28,80] Interestingly, only two mechanical phases were identified at site 1 rather than three distinct clusters as in other sites, as shown in Figure 9b. This demonstrates the inherent heterogeneity of shale. The tested area in the indentation was limited and might not have all the dominant mineral phases. For the same reason, the proportion of the clusters is not exactly consistent with the content weight fraction in XRD results. The inherent heterogeneity of shale means that the minerals distributed in the indentation area vary greatly with the locations. Nanoindentation tests only zoom in to a local site and may not represent the entire mineralogy. For direct evidence of the mineral types in the indentation area, an in situ high imaging technique such as energy-dispersive X-ray spectroscopy (EDS) in conjunction with SEM is recommended for future tests on each indent.

The high heterogeneity of shale poses a challenge to the theoretical modeling and prediction of its mechanical properties on multiscales. The experimental results from nanoindentation tests provide data to the multiscale modeling of shale and correlate the mineral composition to the micromechanical behavior of shale. The strong heterogeneity of the elastic properties observed in nanoscale should be considered

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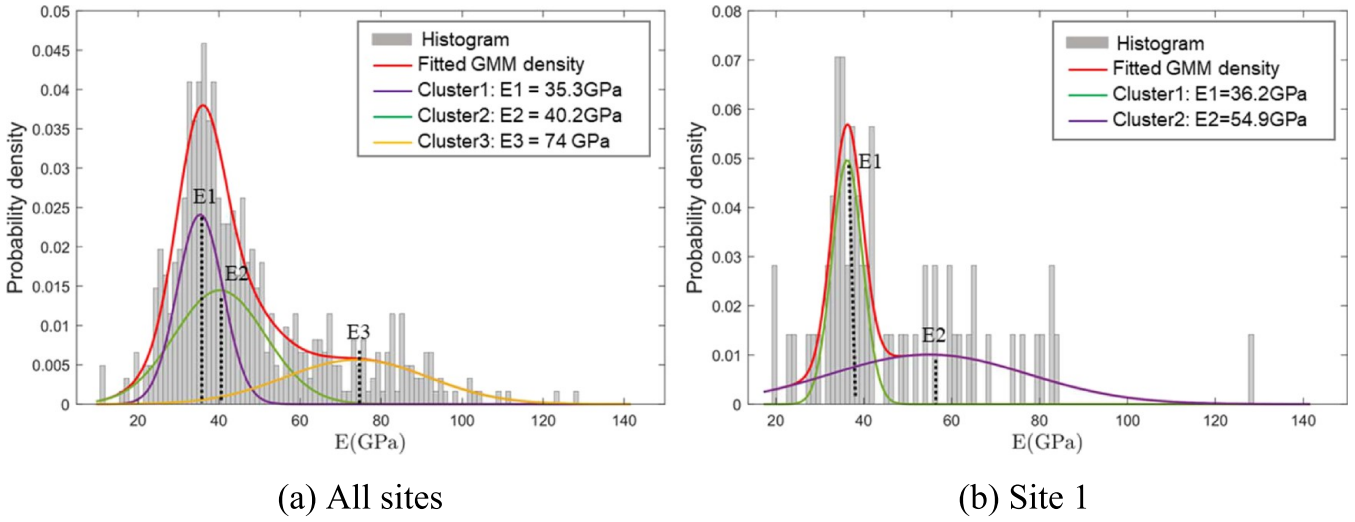


Figure 9. Statistical analysis of the elastic modulus. (a) Gaussian mixture distribution of the elastic moduli of 512 experimental indents for the eight sites. Three clusters were recognized, with mean values of 35.3, 40.2, and 74 GPa, corresponding to the three mineralogical phases of shale: low-, medium-, and high-strength minerals, respectively. (b) Gaussian mixture distribution of the elastic modulus of site 1. Only two clusters were recognized owing to the heterogeneous nature of shale.

Table 2. Statistical Analysis of Elastic Moduli

	mean value (GPa)			mixing proportion		
	cluster 1	cluster 2	cluster 3	cluster 1	cluster 2	cluster 3
total	35.3	40.2	74.0	0.34	0.42	0.25
site 1	36.2	54.9		0.43	0.57	
site 2	34.6	37.7	83.2	0.62	0.30	0.08
site 3	30.9	44.3	86.5	0.47	0.44	0.09
site 4	36.7	53.9	89.5	0.54	0.30	0.16
site 5	31.5	47.4	81.5	0.37	0.37	0.26
site 6	35.0	53.1	85.5	0.50	0.37	0.13
site 7	39.9	66.4	81.4	0.67	0.13	0.30
site 8	34.1	45.9	70.0	0.56	0.13	0.31

into the definition of digital microstructure modeling, which can improve the authenticity and representativeness of the microscopic mechanical model of shale.

4.2. Evolution of Nanoindent Deformation and Associated Energy. The mineral groups are different in the mechanical responses to the indentation load, which manifest themselves as the different pattern in load−displacement curves. It was observed that the load−displacement curves exhibit three patterns. The patterns vary with the maximum indentation depth, which depends on the surface hardness under the probe. Curves that have shallow indents demonstrated a smooth and steep loading process followed by a very short holding creep and the unloading process. Different from low-depth curves, abnormal changes in slope with a sudden increase in the depth were frequently observed in curves with deep indents, usually accompanied with a long creep in the holding process. Based on these observations, we divided the load−displacement curves into three groups according to the different patterns on the curve shapes, as shown in Figure 11a.

To figure out the link between the patterns and the mineralogy of the sample, the associated energy induced in the indentation test was compared. The hysteresis loops enclosed by the loading−unloading curves, representing the energy lost due to plastic deformation and elastic strain owing to residual stresses, varied significantly with depth. This indicates that

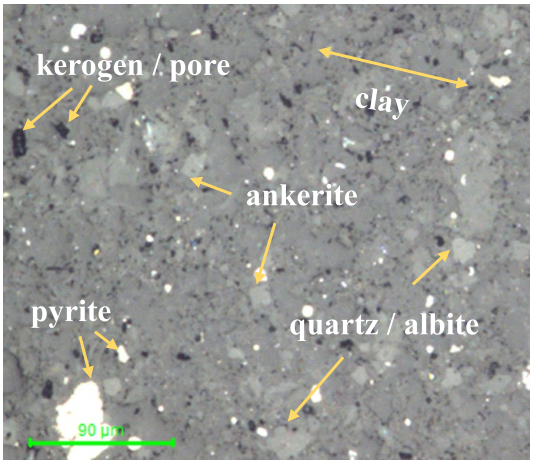


Figure 10. The surface morphology of the shale sample under an optical microscope. The matrix minerals can be identified by their distinct morphological features. (a) Pyrite was easy to distinguish owing to its bright reflection under the optical microscope. (b) Ankerite was the main authigenic carbonate in the shale sample. It was identified by its rhomb shape with a plain surface. (c) Quartz and albite generally exhibited various irregular shapes with sharp edges and were very similar under the optical microscope. Higher-resolution imaging tools or EDS is recommended for distinguishing between quartz and albite. (d) The black area represents either kerogen or pores, and further examination is needed to verify it.

elastic−plastic properties differ between indents. During the indentation process, the total energy U_t, which is the energy of the indenter to the rock by elasticity and plasticity, can be determined by the area under loading and holding curves. As shown in Figure 11b, the total energy consists of two parts: the total irreversible energy U_{ir} and elastic energy U_e. The irreversible energy U_{ir} is composed of pure plastic energy U_{pp} that was lost due to pure plastic deformation and induced crack energy U_{crack}. Several researchers have previously explored the correlation between U_{pp}/U_t and h_p/h_{max}. A general mathematical relationship between U_{pp}/U_t and h_p/h_{max} was observed, which is calculated by the following equation:[82]

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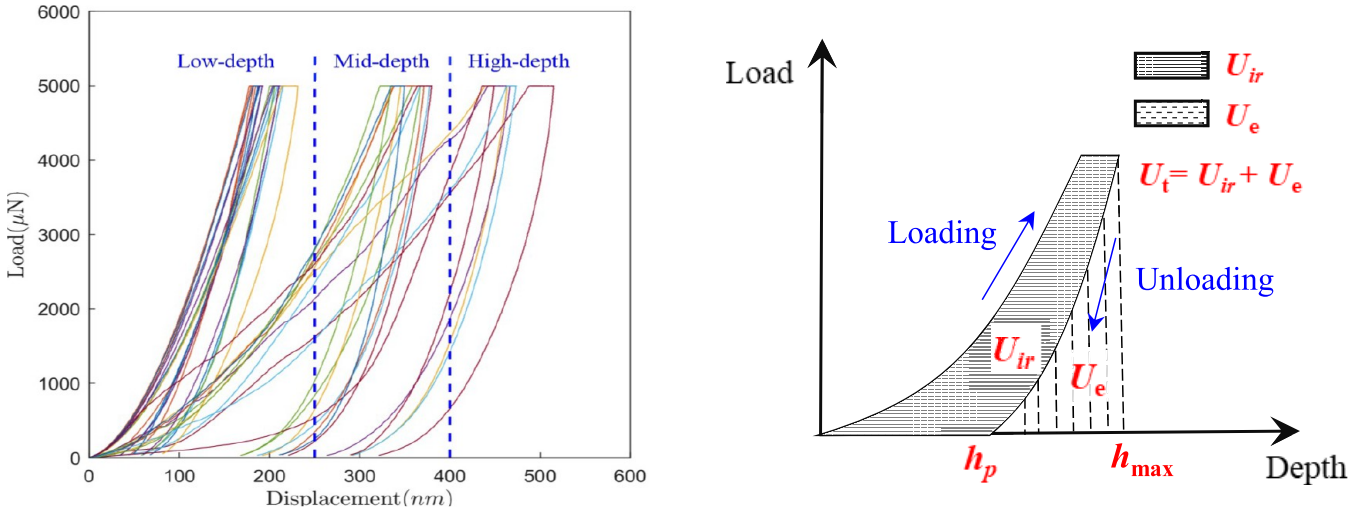


Figure 11. (a) Three types of load−displacement curves with different maximum depths. (b) Schematic of the associated energy induced in the indentation process.

$$\frac{U_{pp}}{U_t} = 1 - \frac{1 - 3\left(\frac{h_p}{h_{max}}\right)^2 + 2\left(\frac{h_p}{h_{max}}\right)^3}{1 - \left(\frac{h_p}{h_{max}}\right)^2} \quad (4)$$

where h_{max} is the maximum depth and h_p is the residual depth after probe withdrawal from the specimen surface. This relationship is explicitly independent of indenter geometry and the materials' properties. During this process, the crack length does not need to be known. Equation 4 was later applied to an energy analysis method for the determination of fracture toughness of shale rocks.[83] Equation 4 does not take the holding process into consideration, and this may cause an underestimation of the plastic work. However, a previous study demonstrated that adding the holding process to the calculation only makes a small difference and the value does not change too much.[84]

The mechanical properties and the related energy of the groups were summarized in Table 3. The hysteresis loops enclosed by the load−displacement curves at low depth were narrow, with less plastic work, indicating that the elastic deformation was dominant in the high-strength clusters. The plastic works of the deep load−displacement curves occupied a significant proportion of the indentation process, clearly larger than the other groups owing to the massive plastic work of the plate- and sheet-like clay minerals. The average plastic work ratios of low-, mid-, and high-depth curves at the indentation sites are shown in Figure 12. Despite the heterogeneity of shale, the average plastic work ratios corresponding to the three groups of curves were in good agreement across all indentation sites.

Mineralogy is a vital component controlling the mechanical behaviors of shale. The low-depth curves showed that minerals with high hardness, such as quartz, had the lowest pure plastic work ratio of 0.06−0.18. For minerals with medium hardness, the pure plastic work was prominent, around 0.35 of the total work. The high-depth curves corresponded to sheet- or plate-like clay minerals and organic matters and showed the highest plastic work ratio, as high as 0.53. This means that more energy was dissipated owing to the cohesive frictional behavior between particles. The elastic−plastic properties vary among the mineral grains. Minerals with high hardness were highly elastic, exhibited a small irrecoverable deformation in

Table 3. Average of the Mechanical Properties and the Associated Energy

	E (GPa)	H (GPa)	h_{max} (nm)	U_{ir} (pJ)	U_t (pJ)	U_{pp}/U_t
Site 1						
low depth	71.56	9.66	205.38	133.23	414.12	0.11
mid-depth	44.13	2.69	337.43	393.82	648.91	0.36
high depth	32.03	1.40	439.60	579.57	830.22	0.48
Site 2						
low depth	70.33	8.72	213.17	147.76	429.85	0.14
mid-depth	38.89	2.78	344.87	398.29	681.57	0.34
high depth	28.95	1.16	540.85	646.13	901.05	0.53
Site 3						
low depth	89.37	13.76	177.11	60.08	340.42	0.06
mid-depth	41.90	2.70	338.28	400.48	669.86	0.36
high depth	31.45	1.37	444.38	671.07	925.53	0.47
Site 4						
low depth	79.39	9.59	203.56	137.41	400.81	0.15
mid-depth	42.32	2.58	341.25	406.42	664.42	0.38
high depth	33.10	1.43	429.45	623.93	874.20	0.46
Site 5						
low depth	79.61	10.07	204.66	138.85	395.87	0.18
mid-depth	41.95	2.65	342.58	438.01	702.70	0.36
high depth	34.59	1.38	436.96	629.16	870.91	0.50
Site 6						
low depth	71.51	8.88	211.71	136.25	413.43	0.16
mid-depth	42.32	2.45	347.64	424.53	678.41	0.37
high depth	30.02	1.28	453.46	696.70	952.63	0.48
Site 7						
low depth	73.25	11.39	201.22	92.80	399.51	0.09
mid-depth	49.54	3.08	316.35	380.11	629.47	0.36
high depth	34.89	1.35	442.01	664.28	897.61	0.52
Site 8						
low depth	78.71	9.63	202.52	124.95	396.12	0.15
mid-depth	43.34	2.99	327.26	346.24	621.63	0.32
high depth	31.81	1.32	447.85	654.78	896.52	0.50

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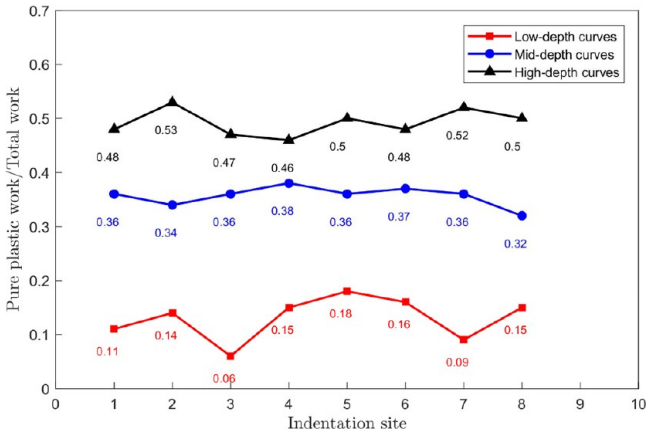


Figure 12. Average plastic work ratios (U_{pp}/U_t) of the low-, mid-, and high-depth curves at the indentation sites. The low-depth curves (red) represent the high-hardness constituents of shale, the mid-depth curves (blue) represent constituents with medium hardness, and the high-depth curves (black) correspond to soft and yielding materials such as clay minerals and organic matter.

resistance to indentation loads, and emerged as the main components of the strength of shale. In contrast, micrometer clay minerals played a dominant role in overall plasticity and induced significant irreversible deformation in reaction to the load. Carbonates, such as ankerite in the shale sample, also formed an important part of the plastic performance of shale. The abnormal jump steps or sudden changes in slope, known as pop-ins, occurred frequently in the load−displacement curves when the minerals being indented exhibited a plastic behavior. This abnormal phenomenon might be directly related to the microstructure of shale, and this is discussed later in the discussion. The relationship between hardness and the plastic work ratio was fitted by using an exponential function, as shown in Figure 13a,c. It was found that hardness had a two-term exponential relationship with total plastic work U_{ir}, as depicted in Figure 13d. With the decrease of hardness, the proportion of plastic work increases exponentially. These fitted relationships are important for predicting the viscoelastic behavior of shale given the hardness of the constituents.

4.3. Spatial Distribution of Mechanical Properties. The nanoindentation tests were conducted in a consistent moving pattern, migrating along the array in a serpentine path from the chosen starting point. With a known array length and spacing between adjacent indents, we matched the mechanical properties derived from the load−displacement curves with the spatial locations on the indentation sites. This spatial distribution mapping provides a way to infer the constituent phases of shale and the skeletal structure of the indented surface. By assembling the tested elastic modulus and hardness in order, we restored and mapped out the distribution of mechanical properties of sites 6, 7, and 8 as shown in Figure 14.

Closed isoline curves with the same mechanical properties divided the map of mechanical properties into subareas, which confirmed the strong heterogeneity of shale in the tested microscale window. Consistent with the surface morphology of the shale sample under an optical microscope, the gathered blocks of different colors imply the location of certain mineral particles. These minerals can be distinguished through a combination of the spatial mapping of their elastic moduli and hardness. Figure 14 shows that the regions of low elastic modulus and hardness occupied a large proportion of the test area, shown as the connected blue zones, and this is in line with the characteristics of the clay matrix, marked as phase 1 in the map. The regions of high elastic modulus and hardness, shown as red and yellow independent blocks, were related to large and hard grained minerals marked as phases 2, 3, and 4 in the figure. These granular minerals were further distinguished with the assistance of the distinct morphological features and mechanical properties of the minerals.

Quartz crystals usually have a large grain size ranging from 15 to 40 μm. They are identified by typical straight and sharp edges, while their shapes may be irregular.[28] Previous studies have reported that the elastic modulus of quartz was in the range of 70−110 GPa and its hardness was in the range of 12−16 GPa.[28,80,85] Thus, we inferred that the red closed area, marked as phase 2, was most likely to feature quartz grains. Feldspar-group minerals are generally columnar or plate-like particles as a consequence of the characteristic crystalline process of formation. Prior studies have suggested that feldspar minerals have elastic moduli in the spectrum of 45−80 GPa and hardness within 9−10 GPa,[28,80,81,85] which are very similar to those of the area marked as phase 4 in Figure 14. For dolomite-group minerals, an approximate rhombohedral shape is commonly observed, and their grain size ranges from 20 to 50 μm with a hardness of 7−8 GPa. Hence, the mineral labeled as phase 3 probably featured ankerite grains.

It must be noted that the distribution mapping based on the elastic modulus and hardness only roughly reflects the spatial location of the minerals. Some minerals with similar mechanical properties, a low proportion, or small sizes may not be recognized on the map. For example, the transition zone between the large grain minerals identified and the clay phases might have represented a mixed matrix phase containing small hard inclusions of different strengths. Nevertheless, the spatial mapping of the mechanical properties shows the tremendous heterogeneity of shale and the differences in mechanical properties of the mineral phases. To confirm the mineral distributions inferred in this study, energy-dispersive X-ray spectroscopy (EDS) in conjunction with SEM is recommended for future tests on each indent.

4.4. Influence of Water Vapor on Mechanical Properties. To investigate the alteration in the mechanical properties after water vapor treatment, the shale sample was treated under a constant relative humidity of 85% for up to 20 days. Nanoindentation tests were conducted on the eight sites before and after water sorption treatment. Results showed that the mechanical weakening of the shale sample was less obvious after 10 days of treatment. After 20 days of water vapor sorption, the average elastic modulus and hardness of indentation sites exhibited reductions after 20 days of water sorption over a range of 0.6−15% in the elastic modulus and 8.6−17.8% in hardness. The comparisons of the elastic moduli and hardness of the sample before and after 20 days of water vapor treatment are shown in Figure 15 and Table 4. The indentation moduli and hardness of each test were divided into three mineralogical phases based on their distributions by means of k-means clustering algorithm, as the ellipses with different principal axes show in Figure 16. The principal axes of the ellipses became smaller after water treatment. The points representing the mechanical properties of the three clusters of minerals in shale became more concentrated and closer to the region of smaller elastic modulus and hardness. This phenomenon was especially prominent in low- and mid-

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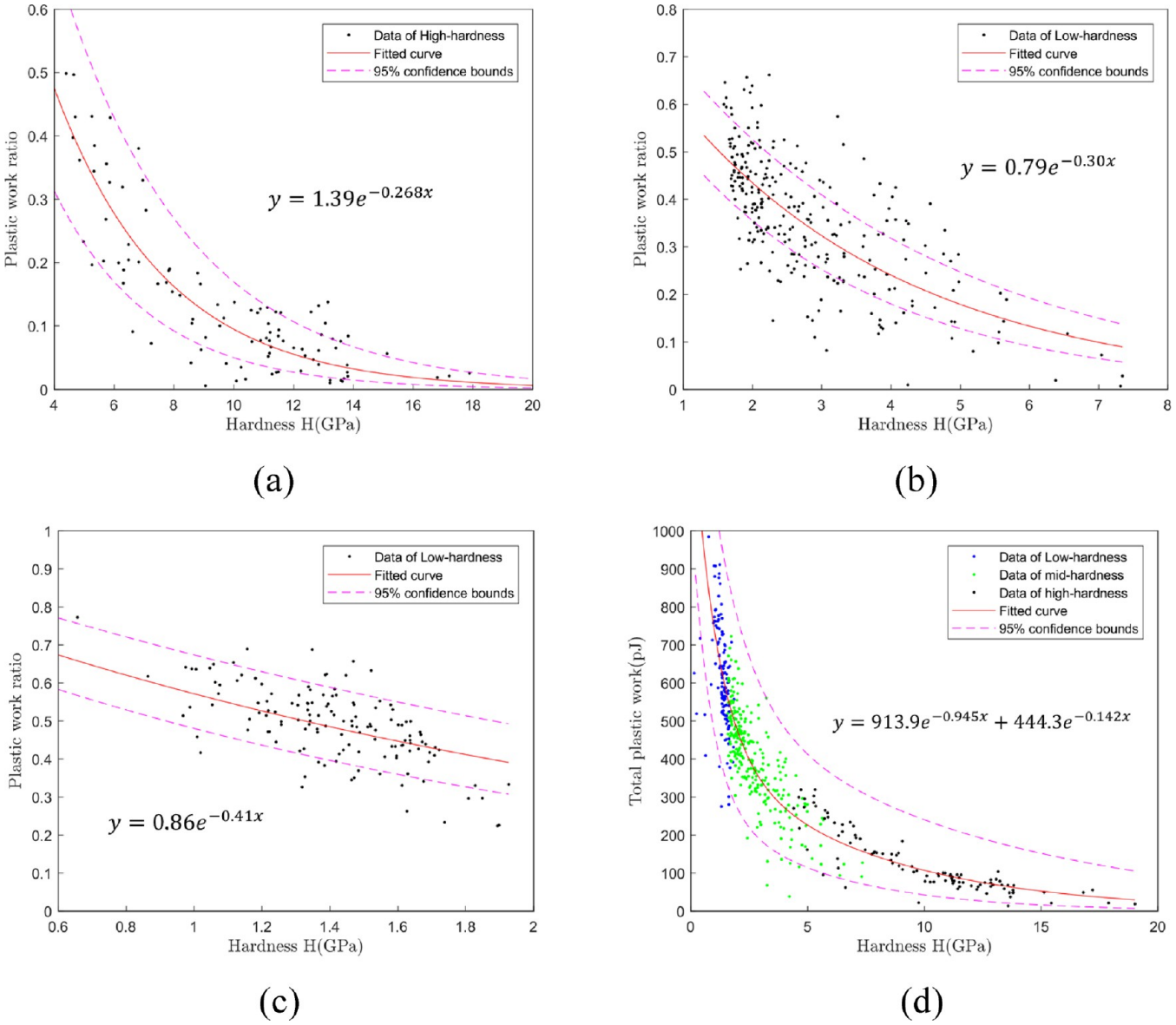


Figure 13. Fitting relationship between statistically deconvoluted hardness, plastic work, and pure plastic ratio for mechanically distinct phases of shale: panels (a), (b), and (c) show the fitting curves of hardness and the plastic work ratio. Panel (d) shows a prominent two-term exponential relationship between hardness and plastic work.

strength minerals, indicating that the alteration had mainly occurred in the two clusters. The alteration was found to not be apparent in clusters of the high-strength minerals.

There are several possible explanations for the mechanical changes of the shale sample. First, it may be attributed to the dissolution of diagenetic minerals. XRD analysis showed that the shale sample contained significant amounts of pyrite with a weight fraction of 3.3%. Pyrite can react with water to dissolve in oxygen and form sulfate and iron oxyhydroxides. Previous work has shown that pyrite dissolves easily even under anaerobic conditions.[86] The pyrite−water reaction facilitated other diagenetic minerals, such as carbonate and clay minerals, which were softened or dissolved during the water sorption process, as shown in Figure 1a. The dissolution of minerals created more pore space that facilitates further water invasion to weaken the cementation between the minerals.

The phyllosilicates, including clay minerals, also played an important role in the weakening behavior of shale due to their layer structure. Chlorite-serpentine and illite are the main rock-forming phyllosilicates in the shale sample. Illite is one of the T-O-T clay minerals with a limited capacity for cation exchange and insignificant hydration swelling compared with montmorillonite or vermiculite. This is because of the sheet arrangement, degrees of isomorphic substitution, and structure of interlayer cations.[72] Illite contains nonexchangeable K^+ as the major interlayer cation that balances the negative charge and, thus, represents a limited hydration effect. Even though hydration did not cause prominent swelling on illite and chlorite-serpentine, it generated sufficient osmotic pressure to cause the separation of the phyllosilicate platelets that easily dispersed and migrated into pores under the influence of the flowing fluid.[87] In addition, a previous study showed that the deteriorating effect of water significantly reduced the fracture toughness to allow fracture propagation under lower stress intensity.[88] This contributes to the propagation of microcracks

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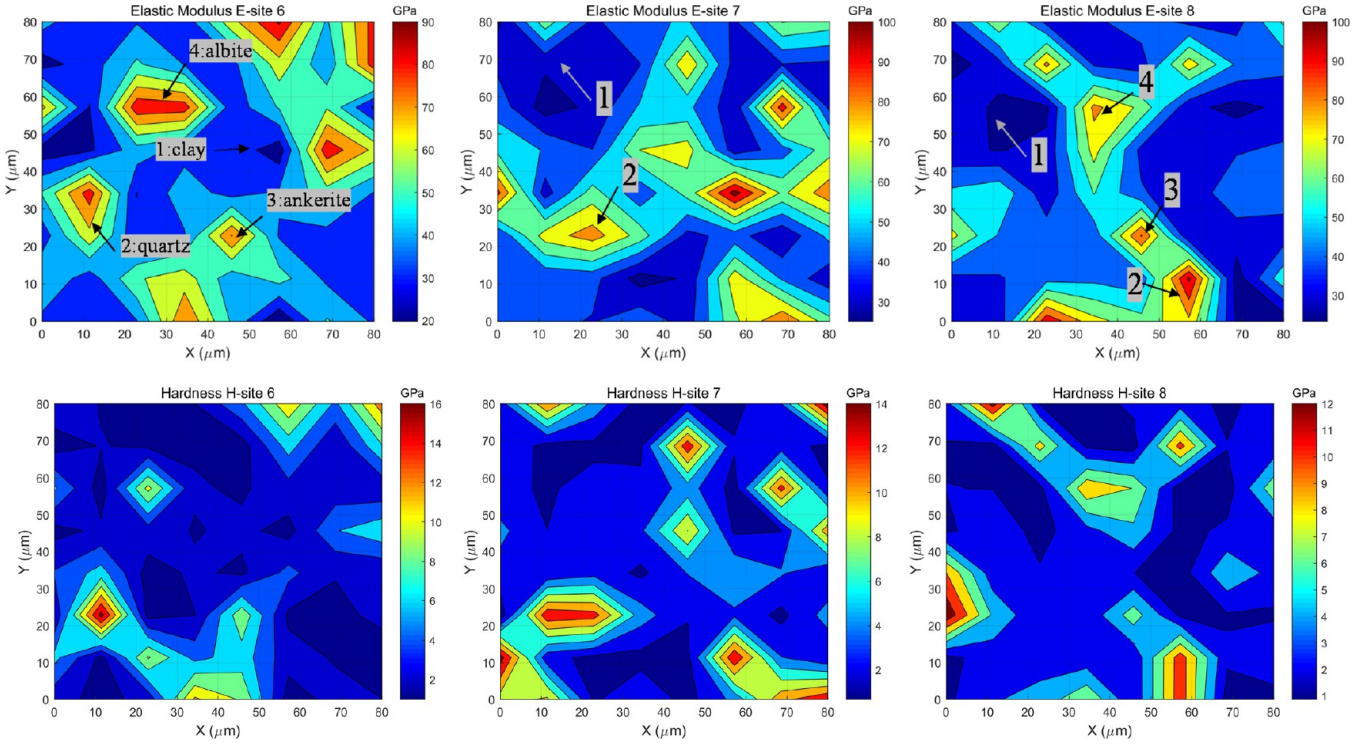


Figure 14. Spatial mapping of the elastic modulus and hardness of the indentation area. The constituent phases of shale and the skeletal structure of the indented surface can be inferred by mapping the spatial distribution of the elastic modulus and hardness combined with the distinct morphological features of the mineral grains.

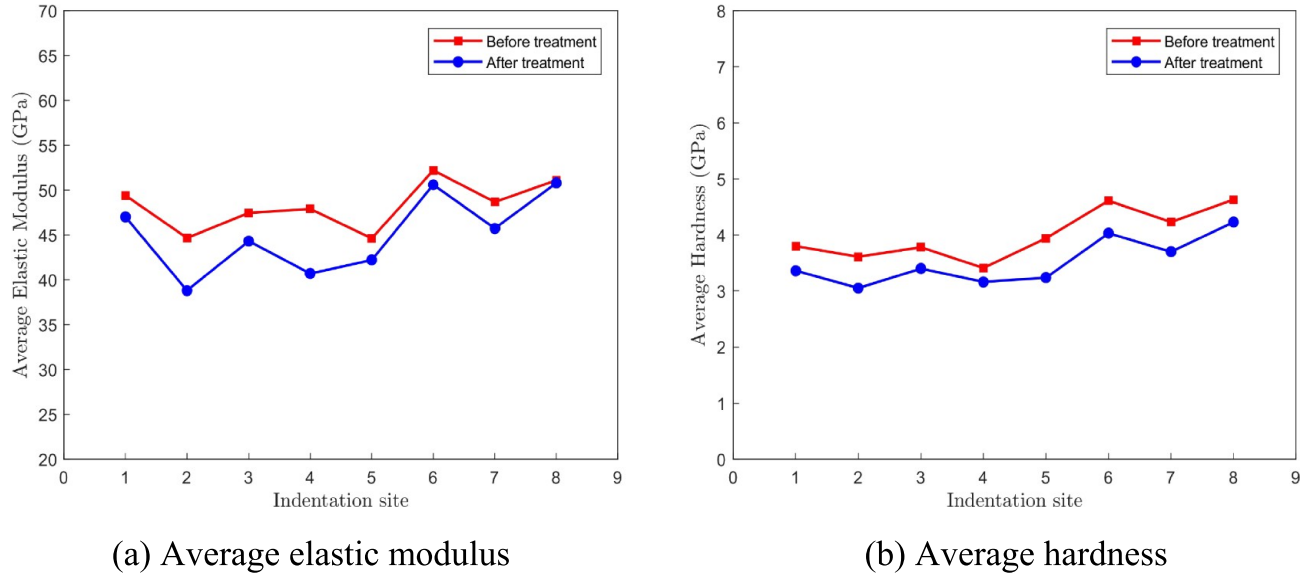


Figure 15. Mechanical properties of the shale sample before and after water treatment.

resulting from the nonuniform swelling of minerals and leads to strength and stiffness reduction and, ultimately, a failure.

In our experiment, a slight increase of the elastic modulus at the beginning of the water treatment was observed, which agreed with the results of previous experiments.[27] This may be caused by capillary suction. At low saturation, water filled the small pores and narrow spaces between grains, generating high capillary suction inside the shale that acted as a confining pressure and led to the closure of pre-existing microcracks—possibly increasing the strength after water vapor sorption.[73]

5. DISCUSSION

The shapes of the load−displacement curves not only can be used to extract the mechanical properties but also contain rich information on the structure of the material. As mentioned before, abnormal changes in slope, like pop-in or elbow events, are commonly observed in load−displacement curves. Some studies have offered possible interpretations for these phenomena. One popular explanation of pop-in events in bulk metallic glass and metal alloys is the negative sensitivity to the strain rate originating from dynamic strain aging, whereby solute atoms diffuse around dislocations and retard mo-

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Table 4. Elastic Moduli and Hardness of the Shale Sample before and after Water Treatment

	average elastic modulus (GPa)			average hardness (GPa)		
	before	after	reduction	before	after	reduction
site 1	49.4	47.0	4.8%	3.80	3.36	11.6%
site 2	44.7	38.8	13.2%	3.61	3.05	15.5%
site 3	47.5	44.2	6.9%	3.78	3.40	10.1%
site 4	47.9	40.7	15.0%	3.41	3.16	7.3%
site 5	44.2	42.2	4.5%	3.94	3.24	17.8%
site 6	52.2	50.6	3.1%	4.61	4.03	12.6%
site 7	48.7	45.7	6.2%	4.23	3.70	12.5%
site 8	51.1	50.8	0.6%	4.63	4.23	8.6%

tion.[89,90] In cases of indenter uplift, the material expansion during the slow transformation to the amorphous phase (for instance, the amorphization of silicon upon pressure release) results in a sudden volume release and leads to the uplift of the material surrounding an indenter and hence change in the unloading curve.[83,89,91] A detailed discussion of these explanations for sudden changes in the slope can be found in the aforementioned references. As a strongly heterogeneous multimineral aggregate and a natural material, shale has unique characteristics. It is necessary to interpret the pop-in phenomena from the perspective of the microstructure of shale.

The load−displacement curves can reflect the mechanical response of a specific local volume below the probe. In direct contact with the tip of the indenter are the surface minerals that can be either hard or soft. Previous studies have indicated that the microstructure beneath the contact mineral, which acts as the substrate, has a significant influence on the results of nanoindentation, especially heterogeneous materials like shale. It contains cracks at the nano- and microscales with a complex pore architecture, as shown in Figure 17. Although the total crack porosity may be small, these pores and cracks have a significant impact on the mechanical and transport-related properties of shale.

Similar to the stress−strain curves of rock, the load−displacement curves were divided into several stages as shown in Figure 17c. The first stage is nonlinear elastic loading, where the curvature of the curve is roughly quadratic. In this stage, the nanocracks and some crack-like pores that were very thin in one direction gradually closed under pressure, as shown in Figure 17d,e. In the second stage, the load−displacement curve underwent an elastic−plastic loading process where irreversible changes occurred. Pop-in events generally occurred in this stage. Instead of the ductile and failing stages of the complete stress−strain curve of rocks, a holding stage occurred and involved the creep behavior of the contact mineral followed by a nearly linear unloading process. After this, a permanent residual impression was left on the surface, as shown in Figure 17g.

The resistance of hard materials to the indentation load prevented the indenter tip from going further downward, resulting in less deformation along the direction of indentation. When the load was removed from the surface, the hard minerals regained their original shape and the elastic strains within the minerals were released. If the probe touched the soft constituents, like clay and kerogen that are prone to yielding under a large pressure of the indenter tip, the probe would penetrate the inclusion and might reach the pore structures underneath. This type of deformation manifested itself as jump steps or sudden changes in slope in the load−displacement curve. This is the reason why it was observed that the pop-in events were the most pronounced for soft minerals.

The occurrence of the pop-ins may depend on the shapes of the diversiform pores below the surface. For example, the tip of the indenter penetrated the surface inclusion and reached a silt-like pore, which is very thin in one dimension, and this may trigger a sudden pressure release from the indenter tip. This led to a small creep deformation until the tip had gone through the pore and reached the resistant matrix or inclusion. The sudden small creep deformation was shown as a small pop-in event in the load−displacement curves in Figure 17a. Pyriform pores might prolong this creep deformation and caused a larger creep deformation in the loading process, namely, a step-shaped pop-in event as shown in Figure 17a. For irregularly shaped pores like the U-shape pore shown in Figure 17b, the indenter tip underwent two or more pressure releases in a row, which might have caused serial pop-in events.

The discontinuities on the sample surface may be another reason for pop-in events, which may be triggered as a consequence of (1) the generation of microcracks in the

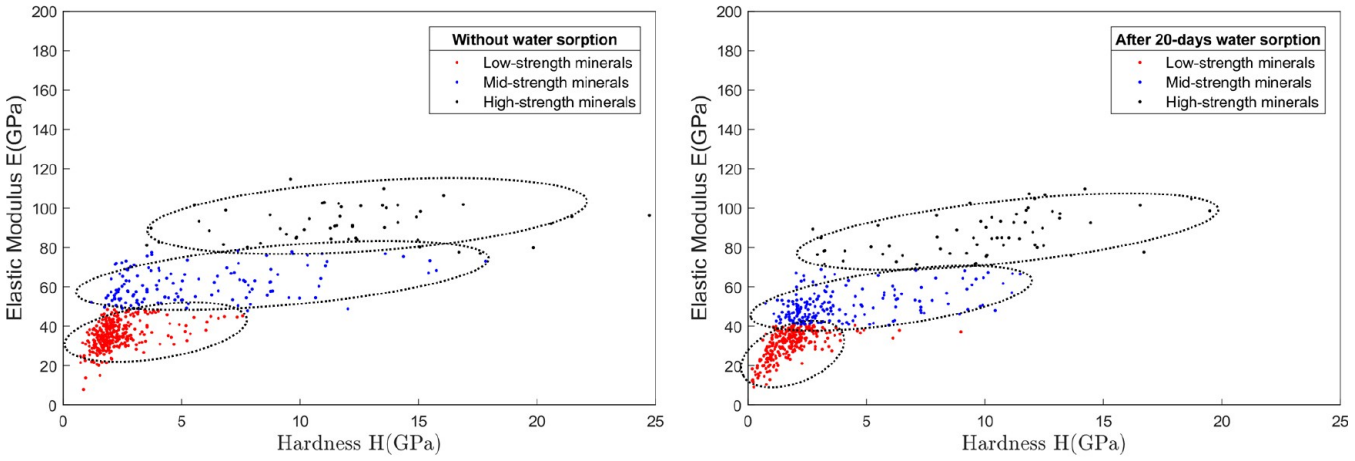


Figure 16. Analysis of the mechanical properties of the indents before and after water vapor sorption via the k-means clustering algorithm.

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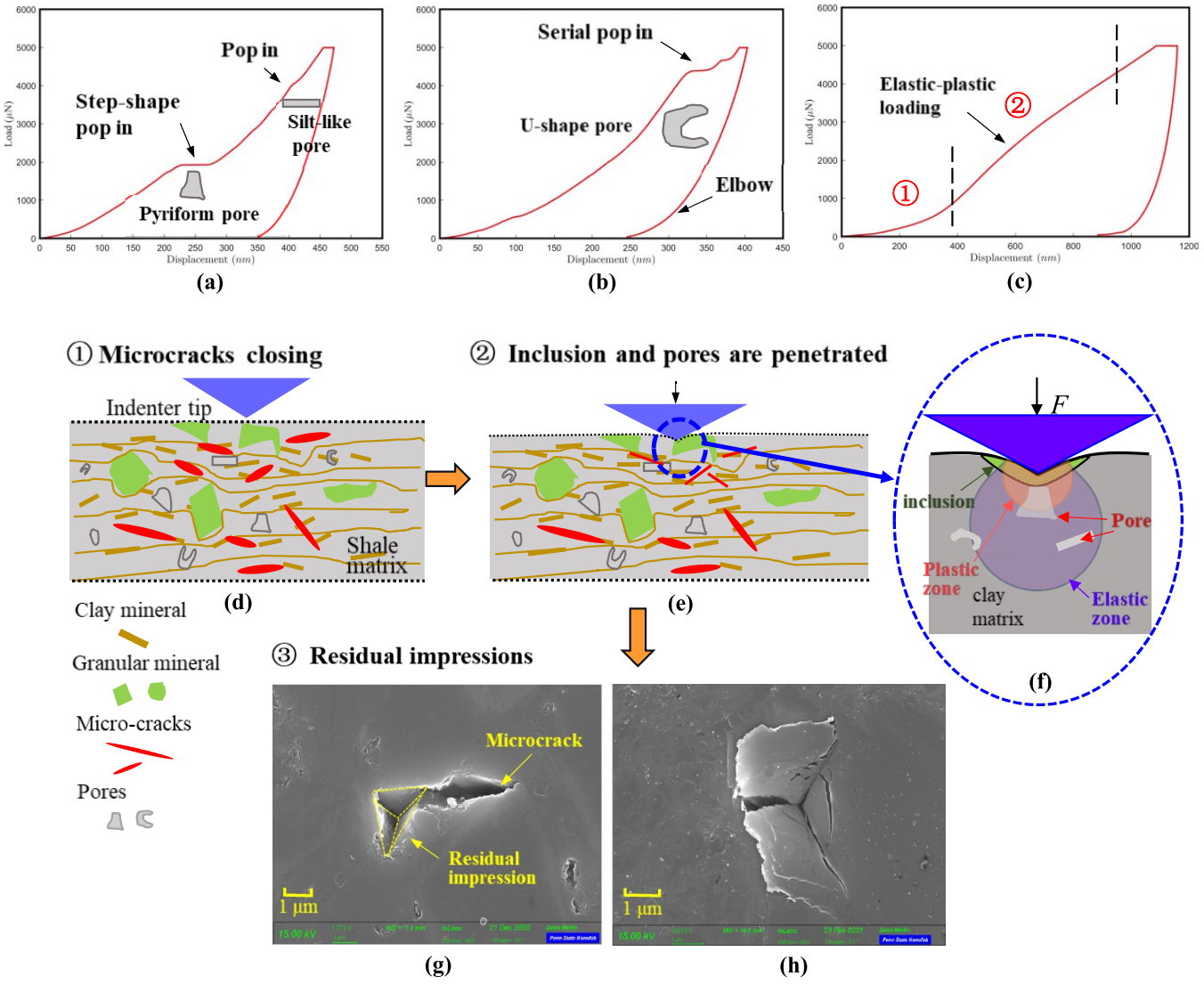


Figure 17. Schematic of the influence of the microstructure of shale on the load−displacement curves. The microstructures in the substrate matrix, pores, and cracks impact the shape of the load−displacement curves. Panels (a), (b), and (f) show the abnormal changes in slope, such as pop-in events, owing to the sudden pressure release on the tip of the indenter when it penetrates the surface inclusion and reaches pores of different shapes below the inclusion. Panels (c−g) summarize the stages of the indentation process: (1) nanocracks and crack-like pores close under pressure in the initial nonlinear elastic loading; (2) irreversible deformation occurs in the next elastic−plastic loading process, and the tip continuously penetrates the inclusion and pores; and (3) the tip holds on to the surface at maximum load and then withdraws, leaving a permanent residual impression on it. Panel (h) is an amplification image of the indent under SEM, and it shows that microfractures had been induced around the surface indents.

loading segment, (2) indents that fall on the mineral boundary, or (3) microcracks on the sample surface, as shown in Figure 17g. As a brittle material, shale has a high chance of fracturing under the loading process during the indentation. The amplification image of the indent under SEM verified that microfractures had been induced around the surface indents, as shown in Figure 17h, which were another major reason for pop-in events. Shale rock is an elastic−plastic material, and plastic deformation occurs in the contact area beneath the indenter when the elastic energy exceeds a critical value with the increasing load. The plastic zone was observed to be a semispherical area connected to a surface outside the radius of the circle of contact, as shown in Figure 17f.[75] While the elastic energy remained nearly constant within the localized plastic deformation zones, areas outside it were in a state of nascent plastic deformation and had less elastic energy on account of the increase in contact depth. This energy difference in the deformation zones could have caused crack initiation and plastic deformation near the indenter, accompanied by the piling up around the indents, and resulted in pop-in events.

Although the abnormal changes in the load−displacement curves vary in different materials, there is a consensus that the manifestation of unusual deformation is related to the loading rate; compositions of the material; and, in some materials, the orientation of the grains with respect to the direction of indentation. As for porous materials like rock, the pop-in events are closely related to the microstructure of the tested material. Future work using indent-specified imaging (SEM-EDS) is recommended to confirm the leading parameters influencing these behaviors.

6. SUMMARY AND CONCLUSIONS

In this study, the nanomechanical properties of shale were characterized using the nanoindentation technique. The

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advanced accelerated property mapping mode was used to obtain the spatial distribution of the properties of shale. Statistical methods such as the Gaussian mixture mode and k-means algorithm were used to analyze the elastic modulus and hardness extracted from load−displacement curves. Based on the results, the following conclusions can be drawn:

(1) The micromechanics of shale are conditioned by the mineral composition and its distribution. Three clusters of different mineral phases were recognized through nanoindentation curves. The average value of the elastic moduli of the three clusters ranged from 30.9 to 39.9, 37.7 to 66.4, and 70 to 89.5 GPa, respectively.

(2) The plastic work induced in the nanoindentation process was different between the three mineral clusters with a plastic work ratio at 0.06−0.18, 0.32−0.38, and 0.46−0.53, respectively. A two-term exponential relationship was observed between the hardness and the plastic work.

(3) It is evident that the shale was highly heterogeneous in the nano-to-micron scales based on the mapping of the spatial distribution of its elastic modulus and hardness.

(4) The elastic modulus of shale exhibited a reduction of 0.6−15% after 20 days of water treatment, while its hardness decreased by 8.6−17.8%. The mechanical weakening may be attributed to the dissolution of diagenetic minerals, such as pyrite and carbonates, in water as well as the dispersion and migration of clay minerals owing to osmotic pressure.

(5) The occurrences of pop-in events are related to the microstructure of shale below the tip of the indenter. The microstructure should be further studied to identify the role of the local structure on the micromechanical properties of shale.

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Notes

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